Quantification: 543-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	12
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 12)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2027.92	2066	537.141	px^2
major_axis_length_px	lacuna shape	81.5016	86.4661	16.9465	px
minor_axis_length_px	lacuna shape	33.8846	32.3799	6.9212	px
aspect_ratio	lacuna shape	2.4727	2.3986	0.6666	unitless
eccentricity	lacuna shape	0.8942	0.9087	0.0558	unitless
solidity	lacuna shape	0.8887	0.9115	0.0668	unitless
roots_count	headline	9	9	2.6629	count
ring_length_r30_px	headline	410.167	386.5	87.3632	px
ring_length_r60_px	reported	1111.25	1113.5	218.065	px
owned_length_px	ownership-dependent	3601.47	3557.2	1502.1	px
edge_count	ownership-dependent	169.167	165	69.2017	count
mean_edge_length_px	ownership-dependent	21.1495	21.0104	1.7194	px
perimeter_px	normalised	222.101	229.693	45.067	px
ring_area_r30_px2	normalised	8742.17	8905.5	1127.78	px^2
ring_area_r60_px2	normalised	22189.9	23189	2825.36	px^2
in_frame_fraction_r30	normalised	0.9965	1	0.0123	unitless
in_frame_fraction_r60	normalised	0.9814	1	0.0596	unitless
ring_density_r30	normalised	0.046754	0.0458883	0.00611617	px^-1
ring_density_r60	normalised	0.0498661	0.051119	0.00574932	px^-1
roots_per_100px_perimeter	normalised	4.0113	3.9619	0.6761	per 100 px
ring_attached_length_r30_px	network v2	367.417	351	89.7962	px
ring_attached_length_r60_px	network v2	1006.42	973.5	223.181	px
ring_length_w_r30_px	network v2	463.474	438.055	101.083	px
ring_length_w_r60_px	network v2	1269.36	1267.17	250.919	px
sholl_crossings_r10	network v2	10.8333	11	2.6572	count
sholl_crossings_r20	network v2	13.75	12.5	3.0189	count
sholl_crossings_r30	network v2	14.9167	15	3.4234	count
loop_count	loops, report only	19.9167	18.5	10.1843	count
loop_edge_share	loops, report only	0.4407	0.4525	0.0885	share
ring_width_mean_r30_px	width	7.3596	7.2923	0.3045	px

Per field
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measure	value	units
analysed_area_px2	1.02424e+06	px^2
total_skeleton_length_px	44739	px
canalicular_length_density_per_px	0.0436802	px^-1
junction_count	1155	count
junction_density_per_px2	0.00112766	px^-2
lacuna_count	12	count

lacunae_per_px2	1.172e-05	px^-2
skeleton_component_count	148	count
mean_component_length_px	302.291	px
median_component_length_px	19.5	px
field_length_density_w_per_px	0.0497465	px^-1
field_density_without_flagged_per_px	0.0436802	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	416	px
unattached_component_count	24	count
canalicular_length_density_attached_per_px	0.0436802	px^-1
field_length_density_with_unattached_per_px	0.0440863	px^-1
loop_count	285	count
loop_edge_share	0.4985	share
edge_count	2297	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.9928	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	44739	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2494	90.6112	36.7713	2.4642	0.914	0.9096	0.4535	88.1403	271.181	False
2	3004	86.0791	52.8366	1.6292	0.7895	0.772	0.3181	207.791	919.033	False
3	1698	63.4601	34.3535	1.8473	0.8408	0.955	0.5365	257.425	340.805	False
4	1631	87.5017	30.956	2.8266	0.9353	0.741	0.0248	297.032	38.7695	False
5	1204	55.3793	28.6478	1.9331	0.8558	0.9177	0.1008	394.263	364.613	False
6	2066	81.299	34.8468	2.333	0.9035	0.8619	0.1545	472.247	745.532	False
7	1233	57.0332	27.8479	2.048	0.8727	0.9278	0.052	459.387	484.83	False
8	2066	100.448	26.9826	3.7227	0.9632	0.9018	0.236	550.262	877.61	False
9	2656	108.838	32.5449	3.3442	0.9542	0.8877	-0.0333	642.811	193.327	False
10	2113	91.9731	30.2288	3.0426	0.9444	0.9135	0.0369	728.295	633.991	False
11	2173	86.8531	32.2149	2.6961	0.9287	0.9358	0.0768	787.166	480.067	False
12	1997	68.5432	38.3841	1.7857	0.8285	0.9402	0.1037	859.482	288.042	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	13	514	1397	4244.9	209	20.3105	7.8391
2	12	614	1493	5619.75	273	20.5852	7.0213
3	8	386	977	2306.85	104	22.1812	7.2817
4	9	382	710	1223.9	71	17.2381	7.0693
5	6	355	961	2569.59	127	20.233	7.1713
6	10	354	1023	1830.28	88	20.7986	7.4527
7	5	301	942	3038.05	124	24.5004	7.1774

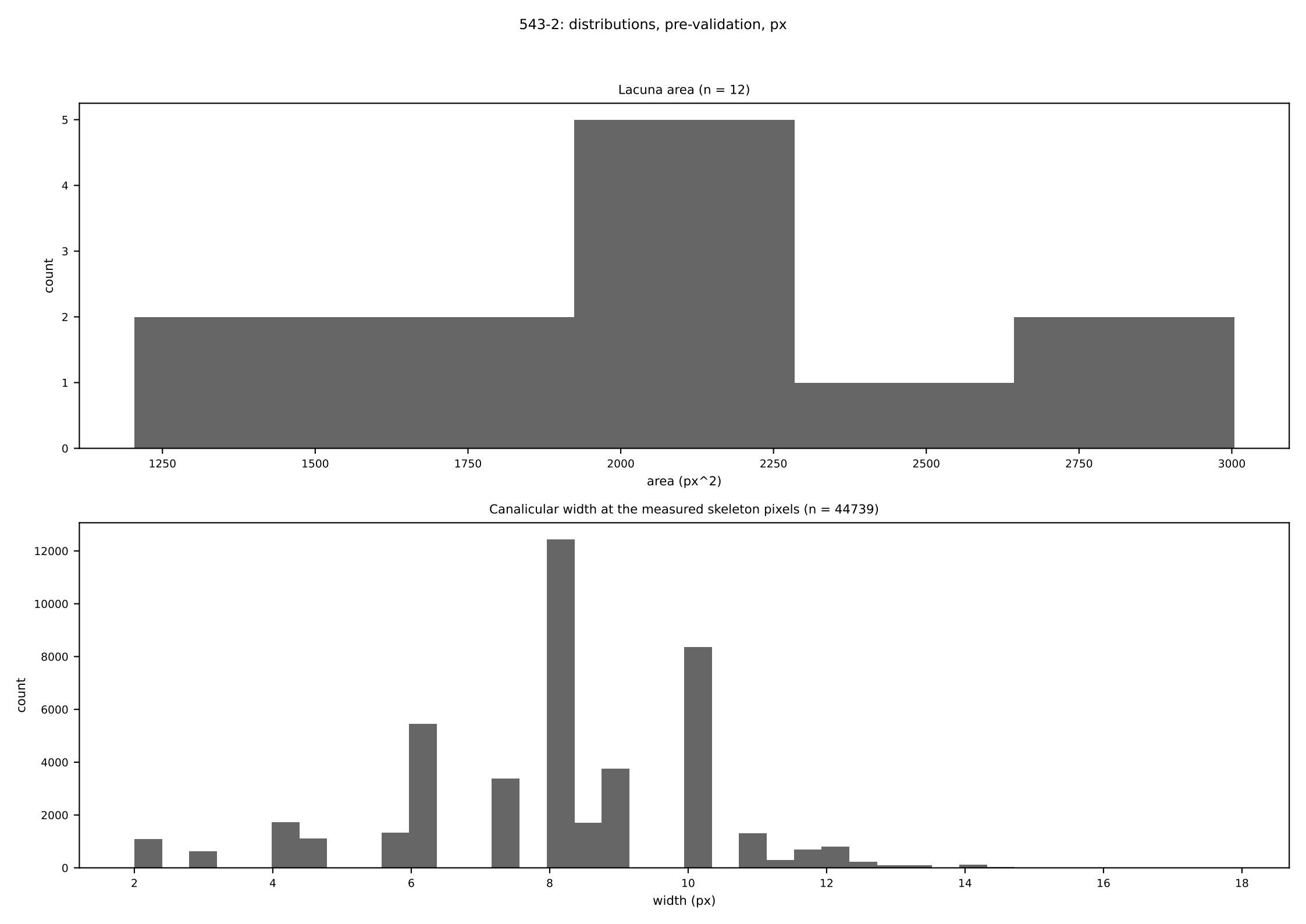
8	12	454	1211	4076.35	190	21.4545	7.5322
9	11	424	1229	5814.11	266	21.8576	7.4156
10	9	387	1204	4944.79	233	21.2223	7.9823
11	7	435	1205	4640.02	205	22.6343	7.0697
12	6	316	983	2909.04	140	20.7789	7.3029

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	244.492	9510	23791	1	0.9843	0.0540484	0.0587197	5.3171
2	295.12	10519	26271	1	1	0.0583706	0.0568307	4.0661
3	172.894	7548	18977	1	1	0.0511394	0.0514834	4.6271
4	232.551	8756	18925	0.9575	0.7928	0.0436272	0.0375165	3.8701
5	154.267	7010	17818	1	1	0.0506419	0.0539342	3.8894
6	231.764	9080	22988	1	1	0.0389868	0.0445015	4.3147
7	155.924	7017	19168	1	1	0.0428958	0.0491444	3.2067
8	254.25	9429	23720	1	1	0.0481493	0.051054	4.7198
9	277.078	10094	25807	1	1	0.0420051	0.0476227	3.97
10	227.622	8939	23523	1	1	0.0432934	0.0511839	3.9539
11	225.522	8872	23390	1	1	0.0490307	0.0515177	3.1039
12	193.723	8132	21901	1	1	0.0388588	0.0448838	3.0972

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	485	1329	587.588	1609.02	13	16	16
2	566	1438	696.63	1702.3	15	21	21
3	354	803	436.505	1121.52	12	14	14
4	356	647	429.534	793.955	12	13	8
5	276	896	396.836	1111.18	8	11	15
6	335	1001	403.706	1172.76	10	11	11
7	251	835	338.108	1073.26	7	11	12
8	437	1067	523.446	1385.47	14	12	17
9	348	893	469.877	1407.8	9	16	15
10	306	1152	439.605	1362.77	8	12	18
11	382	1070	487.09	1361.57	13	16	17
12	313	946	352.764	1130.76	9	12	15


543-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
0
1
2
3
4
5
1250
1500
1750
2000
2250
2500
2750
3000
Canalicular width at the measured skeleton pixels (n = 44739)
count
width (px)
0
2000
4000
6000
8000
10000
12000
2
4
6
8
10
12
14
16
18